Linear programming: solutions

A15-L01 Free variables and minimization

Derivation

Set $u=p-q$, $p, q \geq 0$, and maximize $-3p+3q-v$. Equality gives $p-q-v \leq 2$ and $-p+q+v \leq -2$; the other row is $p-q+v \leq 6$. Slacks are $s_1=2-p+q+v$, $s_2=-2+p-q-v$, $s_3=6-p+q-v$, with all p, q, v, s nonnegative.

Original feasible u maps to $p=\max(u,0)$, $q=\max(-u,0)$. A new feasible point maps back by $u=p-q$, preserving original equality/inequality and reversing the objective sign. Directly, $u=v+2$ and $2v+2 \leq 6$ give $0 \leq v \leq 2$. Objective $3(v+2)+v=6+4v$ is minimized at $v=0$, $u=2$, with value 6. The transformed maximum is -6, attained for example at$(p, q, v)=(2,0,0)$.

Conclusion and checks

Negate the new maximum -6 to return original value 6. This example's optimal u happens to be nonnegative; that does not justify imposing it generally. A feasible problem requiring $u=-1$ would become infeasible.

A15-L02 Complete pivots and a dual certificate

Derivation

Initially $x_3=10-2x_1-x_2$, $x_4=8-x_1-2x_2$, $z=4x_1+3x_2$. Entering x_1 has ratios 5,8, so x_3 leaves. New rows are $x_1=5-x_2/2-x_3/2$, $x_4=3-3x_2/2+x_3/2$, $z=20+x_2-2x_3$.

Enter x_2. Ratios from x_1, x_4 are $5/(1/2)=10$ and $3/(3/2)=2$, so x_4 leaves. The complete dictionary becomes

$$x_2=2+x_3/3-2x_4/3;\ x_1=4-2x_3/3+x_4/3;\ z=22-5x_3/3-2x_4/3.$$

Set $x_3=x_4=0$ to obtain(4,2). Constraints $8+2=10$ and $4+4=8$ are tight; objective $16+6=22$. Dual minimizes $10y_1+8y_2$ subject to $2y_1+y_2 \geq 4$, $y_1+2y_2 \geq 3$, $y \geq 0$. Slack reduced costs give $y=(5/3,2/3)$, both dual equalities, and value $50/3+16/3=22$.

Conclusion and checks

Steps 5,2 lead to optimum(4,2), value 22. Both pivots retain leaving-variable coefficients. The same-value feasible dual independently certifies optimality.

A15-L03 Zero ratios, ties, and symbolic perturbation

Derivation

Entering x_1 gives ratios 0/1=0 for x_3 and 2/1=2 for x_4. Let x_3 leave at $\theta=0$. The dictionary becomes $x_1=-x_3$, $x_4=2+x_3-x_2$, $z=-x_3+x_2$. Decision variables remain 0 with $x_4=2$. Enter x_2:only x_4 has positive denominator, with $\theta=2$. New $x_2=2+x_3-x_4$, $x_1=-x_3$, $z=2-x_4$ has no positive reduced cost. At $x_3=x_4=0$, the optimum is(0,2), value 2. Direct feasibility also forces $x_1=0, 0\leq x_2\leq 2$.

For the symbolic example, ratios are $2+\varepsilon$ and $2+\varepsilon^2$. Their first difference is at ε coefficient 1 versus 0, so$2+\varepsilon^2$ is smaller and x_4 leaves. For positive infinitesimal ε,$\varepsilon^2<\varepsilon$. Afterward $x_3=\varepsilon-\varepsilon^2+x_2+x_4$ has zero constant at nonbasics 0 but strictly positive symbolic value.

Conclusion and checks

A zero step legally changes basis. The first example eventually moves, but its first degenerate pivot itself does not improve the point. Lexicographic comparison begins at low degree; an arbitrary floating ε is not the exact rule.

A15-L04 From Phase I through the original optimum

Derivation

Original slacks $s_1=4-x$, $s_2=-2+x$ give $s_2=-2$ at $x=0$. Auxiliary rows are $s_1=4-x+a$, $s_2=-2+x+a$, $w=-a$. Special a-entry, s_2-exit gives $a=2-x+s_2$, $s_1=6-2x+s_2$, $w=-2+x-s_2$. Point$(x, a, s_1, s_2)=(0,2,6,0)$ is now feasible.

Enter x:ratios 2 from a and 3 from s_1 make a leave. Obtain $x=2+s_2-a$, $s_1=2-s_2+2a$, $w=-a$, with $x=2$, $a=0$, $s_1=2$. Auxiliary optimum is 0. Delete a to get $x=2+s_2$, $s_1=2-s_2$. Restore $z=x=2+s_2$. Enter s_2 and let s_1 leave at $\theta=2$, obtaining $s_2=2-s_1$, $x=4-s_1$, $z=4-s_1$. Thus $x=4$ is optimal.

With bound $x\leq 1$, auxiliary initialization has $s_1=1-x+a$, $s_2=-2+x+a$. Special pivot gives $s_1=3-2x+s_2$, $a=2-x+s_2$, $w=-2+x-s_2$. Ratios 3/2 and 2 make s_1 leave when x enters. Final $x=3/2-s_1/2+s_2/2$, $a=1/2+s_1/2+s_2/2$, $w=-1/2-s_1/2-s_2/2$ proves auxiliary optimum$-1/2$ and original infeasibility.

Conclusion and checks

Distinguish Phase I optimum 0 from original optimum 4. In the changed problem, $x\leq 1$ and$-x\leq -2$ sum to $0\leq -1$. Initial $a=2$ alone never justified an infeasibility conclusion.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

A15-L05 Improving rays and unbounded regions

Derivation

Initially $s=1-x_1+x_2$, $z=2x_1+x_2$. Enter x_1, leave s at $\theta=1$, to get $x_1=1+x_2-s$, $z=2+3x_2-2s$. For entering x_2, the row's subtracted coefficient is -1 and imposes no bound. Set $s=0$, $x_2=\theta$, so $x=(1+\theta,\theta)$.

Choose $x^0=(1,0)$, $r=(1,1)$. Then $r\geq 0$, $A=(1,-1)$ gives $Ar=0\leq 0$, and $c^Tr=3>0$. Every $\theta\geq 0$ is feasible with objective $2+3\theta$, unbounded above. For objective $-x_1-x_2$, nonnegativity gives objective ≤ 0 and the feasible origin attains 0.

Conclusion and checks

Unboundedness of the region is geometric. Objective unboundedness depends on its positive change along a feasible ray, not merely on an infinitely extending drawing.

A15-L06 Preconditions of the potential model

Derivation

Full constraints are $d_s=0$, $d_a\leq -2$, $d_t-d_a\leq 3$, $d_t\leq 4$, $d_v-d_u\leq -1$, $d_u-d_v\leq -1$. The last two sum to $0\leq -2$, making the full LP infeasible because of the disconnected negative cycle.

Restrict to source-reachable{s, a, t}. The path s-a-t gives $d_t\leq 1$. Values $d_a=-2$, $d_t=1$ are feasible, so optimum 1 equals $\min(-2+3,4)$. Adding $d_a\geq 0$ contradicts $d_a\leq -2$ and excludes every correct potential.

Conclusion and checks

Constraints have a scope. An unreachable component's negative cycle does not eliminate a finite s-to-t distance. The repaired potential labels may be negative.

A15-L07 Shared capacities and integrality

Derivation

Use branches s-a-t and s-b-t. For k=1,2, each arc has $g^k\geq 0$, conservation at a, b, and source/sink totals 1. Shared arc capacity is $g^1_e+g^2_e\leq 1$.

A fractional plan sends 1/2 of each commodity on each branch, total 1 per arc. An integral plan sends commodity 1 entirely above and commodity 2 entirely below, total 1 per arc again. Both satisfy the LP, but the first is not single-path per commodity. Allowing splitting does not require it.

With one shared bottleneck of capacity 1 and demands 3/4 each, each commodity separately fits. Combined conservation forces total 3/2 through the bottleneck, violating capacity. Adding independent single-commodity solutions fails because they compete for the same capacity.

Conclusion and checks

This instance having an integral solution does not make every multicommodity LP integral. Integer amounts and unsplittable paths are different extra constraints:an integer demand 2 may split into two paths carrying 1 each and still fail the single-path requirement.

A15-L08 Complementary slackness and theorem boundaries

Derivation

Dual is min $4y_1+2y_2$ with $y_1+y_2 \geq 2$, $y_1 \geq 1$, $y \geq 0$. Point $x=(2,2)$ satisfies $x_1+x_2=4$, $x_1=2$, with value 6. Point $y=(1,1)$ satisfies both dual equalities with value $4+2=6$.

Primal slacks and dual slacks are both$(0,0)$. Thus gap $y^T(b-Ax)+x^T(A^Ty-c)=0+0=0$, certifying both optima. For $x'=(1,3)$, primal slacks are$(0,1)$, objective 5. Gap 1 with the same y is exactly $y_2(2-x_1)=1 \cdot 1$, violating complementarity.

With strict upper constraint $x<1$, every feasible x has a larger feasible$(x+1)/2<1$. No maximum exists, yet objective has finite upper bound 1. The problem is neither infeasible, nor upward-unbounded, nor finitely attained. This does not contradict the theorem, which assumes non-strict LP constraints.

Conclusion and checks

A certificate checks feasibility, objective, and complementary products together. A supremum differs from a maximum attained by a point.

A15-L09 Solution

Set $a=5-x_1-x_2+x_3$, $w=a$. The initial basis $\{a\}$ has a=5 with nonbasics zero. Enter x_1; the unique ratio is 5/1=5, so a leaves and $x_1=5-x_2+x_3-a$. The objective remains w=a; its value at the new basic point is 0. Remove a and restore $z=10-x_2+2x_3$. Enter x_2 and leave x_1: $x_2=5-x_1+x_3$, $z=5+x_1+x_3$. The minimum is 5 at $(0,5)$.

A15-L10 Solution

The dual is $\min\ 8y_1+9y_2, y_1+3y_2\geq 3,\ 2y_1+y_2\geq 2,\ y\geq 0$. For feasible x,y, $c^Tx\leq y^TAx\leq y^Tb$. Moreover $b^Ty-c^Tx=y^T(b-Ax)+x^T(A^Ty-c)$. Every term is nonnegative, so zero gap is equivalent to all complementary products being zero. With $x=(2,3)$ and $y=(3/5,4/5)$, both sets of inequalities are tight and each objective equals 12, proving optimality.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd edition, Chapter 29. MIT 6.046J Fall 2004, Lecture 15; Fall 2005, Lecture 18.

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https://github.com/xinhi6716/computer-science-self-study